MockMind

Train Smart. Interview Smarter.

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Problem Statement

Students fail interviews on communication, not knowledge

Existing mock interview platforms lack personalization, adaptive questioning, and realistic conversational dynamics, making them insufficient for preparing candidates for real-world interviews.

No realistic simulation

Existing tools are glorified question banks with branching logic, not real conversations

AI tools answer for you

No body language feedback

Students never know what they signal non-verbally. Human interviewers notice what nobody tells you.

Pricey platforms

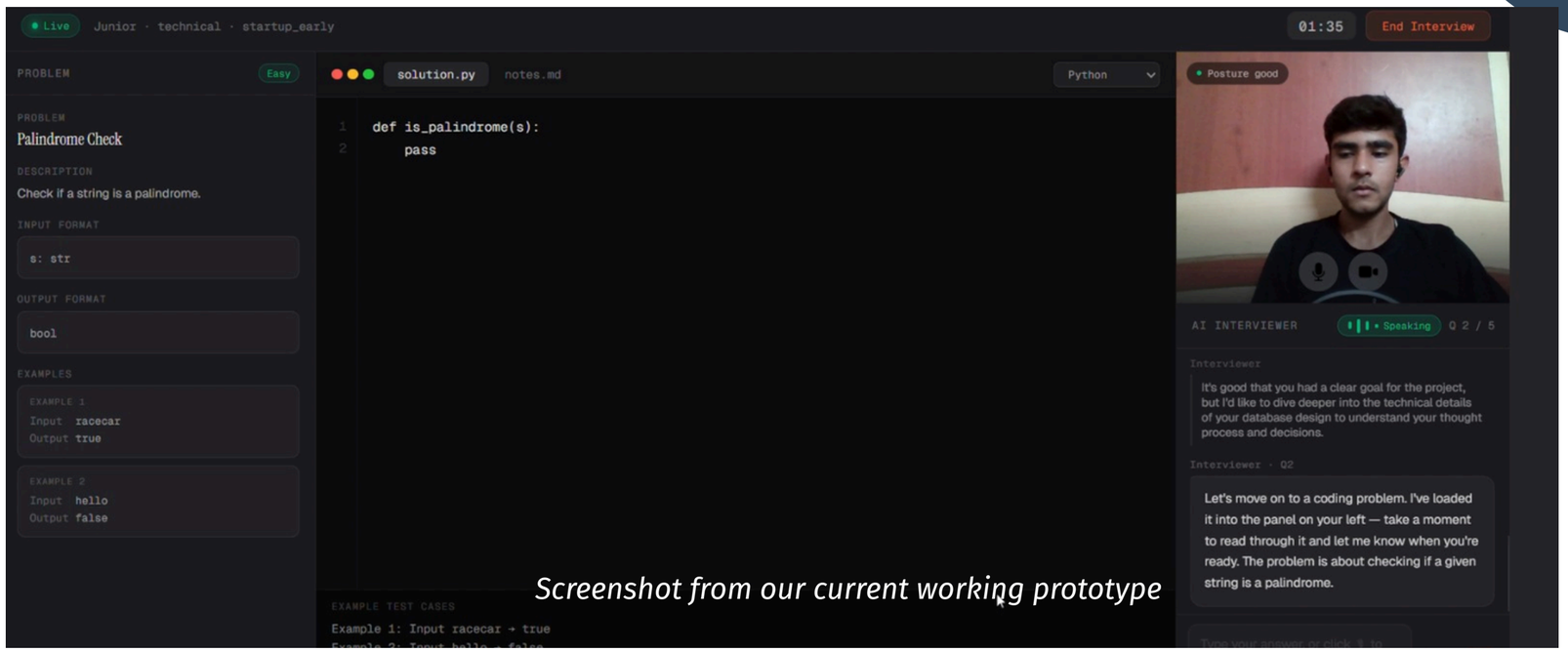
Most students can't afford mock interviews

Extreme latency

Existing tools are too robotic to offer a real simulation necessary for interview prep

Proposed Solution: Agentic AI Interviewer

We propose an Agentic AI-powered mock interviewer that simulates real interview behavior through personalization, dynamic reasoning, and real-time interaction and Post-Interview Feedback & Analytics.



Screenshot from our current working prototype

Dynamic Cross-Questioning

Core USP: Turns interviews into interactive evaluations, with AI asking dynamic follow-up questions.

Post-Interview Feedback & Analytics

Per-question scoring with strengths, weaknesses, and an overall hire verdict

Real-Time Coding + Thought Process Evaluation

Terminal-based code input with
live execution, output validation, and AI evaluation of correctness,
Big-O efficiency,
and explanation clarity

Human-Like Conversational Interaction

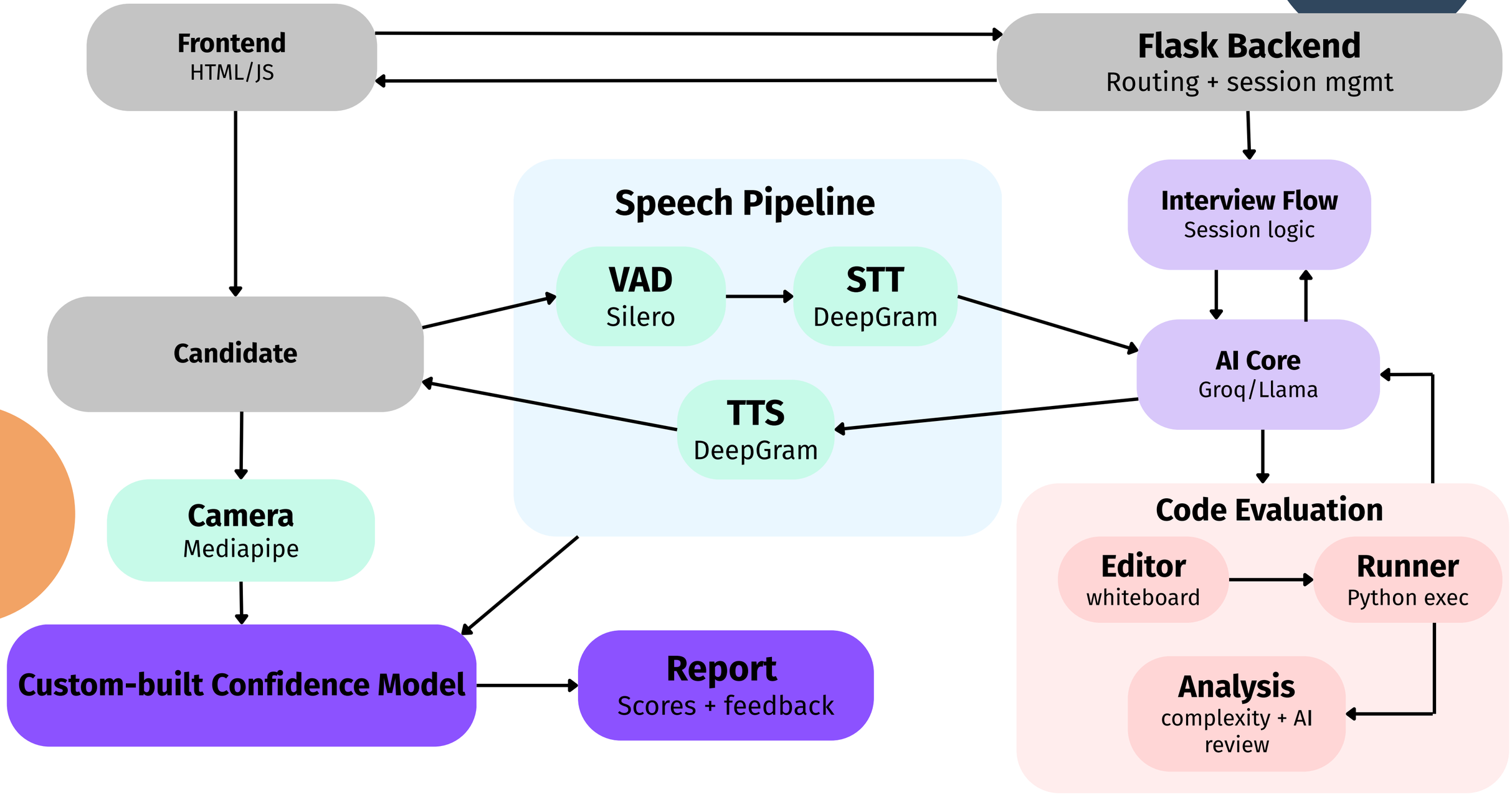
providing a natural human flow a real-time voice-based interaction with Context-aware responses

Personalized Interview Experience

Structured Multi-Round Interview Flow

Seamlessly conducts behavioral and coding rounds in a single session with context-aware evaluation

SYSTEM ARCHITECTURE



Technology Stack Used

Speech (STT/TTS)
DeepGram

AI Core
Groq API (fast LLM inference) via your groq_service

Code Execution
Safe Python sandbox — run_python_code, is_safe_code, analyse_complexity

Backend
Python, Flask, python-dotenv

Frontend
React.js

NLP/Analysis
spaCy for text processing, sounddevice/soundfile for audio

Challenges faced during implementation

Latency
Keeping the full STT → LLM → TTS loop under 600ms for natural conversation feel

Follow-up logic
LLM sometimes returned null for follow-up questions even when score was low — had to add a fallback second Groq call to generate it

Code safety
Running arbitrary user code securely required building a safety checker (is_safe_code) before execution

JSON Reliability
LLM occasionally returned malformed JSON — needed robust try/catch parsing throughout

Empty speech handling
STT sometimes returned empty or [no speech detected] — had to add retry logic to prevent infinite loops

Fluent Talking
Natural conversational transitions between questions make the session feel like a real interview, not a Q&A list

Future Scope and Scalability

Multi-domain support:
HR, case study, system design, and domain-specific interviews (finance, medicine)

Performance tracking
Dashboard showing improvement across multiple mock sessions over time

Interview personas
Choose interviewer style — friendly, aggressive, panel format

3D Avatar
Animate the AI interviewer for a more immersive experience

Scalability
Move to LiveKit for real-time audio infra, Judge0/Piston for scalable code execution in multiple languages

Thank You